Solar Power: A Practical Overview

Test document 1 of 2 for Vectra

1. Overview

Solar power converts sunlight into electricity, either directly with photovoltaic (PV) panels or indirectly by concentrating sunlight to heat a fluid. PV is by far the more common approach. A typical modern panel converts 20 to 23 percent of the sunlight that hits it into electricity, and the figure improves by a fraction of a percentage point every year.

Solar is the fastest-growing source of new electricity capacity worldwide. In most sunny regions it is now the cheapest source of new electricity, with a levelized cost of energy (LCOE) of roughly 30 to 60 US dollars per megawatt-hour for utility-scale projects.

2. How solar plants work

Light striking a silicon cell knocks electrons loose and produces direct current. An inverter then converts that current to alternating current so it can feed the grid. Utility-scale plants arrange thousands of panels in rows, often on trackers that follow the sun and raise annual output by around 20 to 25 percent compared with fixed mounts.

Rooftop systems work the same way at a smaller scale. A typical home system is 5 to 8 kilowatts and can cover most of a household's annual electricity use, with excess power exported to the grid where net metering exists.

3. Cost and economics

Module prices have fallen by about 90 percent over the past decade. Installation, land, grid connection and financing now make up most of a project's cost. Utility-scale LCOE sits at roughly 30 to 60 US dollars per megawatt-hour, and rooftop systems cost more per unit of energy because of their smaller scale.

Solar panels typically last 25 to 30 years and degrade slowly, losing about 0.5 percent of their output each year. The energy used to manufacture a panel is repaid within 1 to 2 years of operation.

4. Reliability and storage

Solar produces nothing at night and less in winter or under heavy cloud. Its capacity factor, meaning the share of the year a plant effectively runs at full power, is typically 15 to 25 percent, depending on location and whether trackers are used. Output peaks around midday, which often does not match evening demand.

Batteries are the main answer. Four-hour lithium-ion batteries are now routinely paired with new solar plants, shifting midday energy into the evening peak. Longer-duration storage is still more expensive.

5. Land use and environment

Utility-scale solar needs about 5 to 7 acres of land per megawatt of capacity. Plants can share land with grazing or pollinator habitat, and panels can also be built on rooftops, car parks and canals, which avoids new land use entirely. Solar produces no emissions while operating, and its lifecycle emissions are roughly 20 to 50 grams of CO2 equivalent per kilowatt-hour.

6. Supply chain and jobs

More than 80 percent of the world's solar module manufacturing is concentrated in a single country, China, which creates supply-chain and trade risks for other regions. Most solar jobs are in installation and project development rather than manufacturing, so they are spread across many local areas.

7. Case study: Harlow Valley Solar Farm (fictional)

The fictional Harlow Valley Solar Farm has a capacity of 320 megawatts and was commissioned in 2021. It uses single-axis trackers and a 4-hour, 100 megawatt battery. It reaches a capacity factor of 27 percent and supplies about 90,000 homes.

Outlook: analysts expect solar to remain the largest source of new capacity additions through the decade, with storage deciding how much of it can be used in the evening.

Note: figures in this document are rounded and illustrative, written as test data.